# Risk Plan - Sprint 2

Project: Factory Inventory & Production Management System (MVP) Role: Risk Manager Sprint: 2 Version: 1.0

## 1. Sprint Context

Sprint 2 focused on core MVP implementation: RBAC authentication, master data, GRN workflow, BOM, production batches, FEFO dispatch, reports, audit logging, and seed data.

## 2. Risk Management Objectives

? Protect inventory accuracy during GRN confirm, batch complete, and dispatch operations.

? Reduce security and access risks caused by role misconfiguration.

? Control integration risk between Firebase Auth, MongoDB, and Express APIs.

? Maintain delivery confidence while implementing many modules in one sprint.

## 3. Key Risks Identified

1. Inventory integrity risk: incorrect stock deductions during batch completion and dispatch.
2. Authorization risk: backend routes accidentally exposed without requireRole checks.
3. Data model consistency risk: mismatch between batch/stock movement references and reports.
4. Scope compression risk: multiple critical modules delivered in parallel can reduce test depth.
5. Seed reliability risk: invalid owner UID/email setup can block first-time onboarding.

## 4. Mitigation Strategy

? Enforce server-side RBAC middleware on all privileged modules.

? Require validation via Zod schemas at body/query boundaries.

? Use MongoDB transactions for multi-collection write operations in GRN, batch, and dispatch flows.

? Track critical actions in AuditLog and StockMovement for post-incident traceability.

? Keep a repeatable seed script and explicit setup instructions in root docs.

## 5. Monitoring and Controls

? Daily risk check during standup against blocked tasks.

? Mid-sprint checkpoint on auth flow and inventory mutation endpoints.

? End-sprint review against integration tests and walkthrough scenarios.

## 6. Sprint 2 Risk Outcome

Core functional risk remained moderate but controlled. Major risk exposures were transferred to Sprint 3 hardening actions (deployment and runtime behavior under Vercel).